Syllabus

Learning Outcomes

The class is aimed at teaching the Lagrangian and Hamiltonian formulation of classical dynamics. It offers a more elegant solution to classical mechanics problems and a different viewpoint, which allows a different perspective on classical mechanics. The application are spacecraft problems.

The class is open to senior undergraduate students and counts as Dynamics and Controls area for them.

Required Course Materials

No book is required; the course script will be made available on Brightspace. Additional, optional resources are discusssed in class.

Topics to be covered:

1. Lagrange Equations
   (a) System identification
   (b) Generalized coordinates
   (c) D'Alambert's Principle and derivation of the Lagrange equations
   (d) Lagrange equations for non-holonomic systems
2. Hamilton's Theory
   (a) Hamilton;s equations
   (b) Hamiltonian principle
   (c) General variational principles
   (d) Phasespace and Liouville's principle
   (e) canonical transformations
   (f) Hamilton-Jacobi theory
3. Non-linear Dynamics
   (a) Dynamical Systems of contraction of the phase-space volume
   (b) Attractors
   (c) Equilibrium solutions
   (d) Bifucations
   (e) Lyapunov exponent and chaos

Grading

40% homework
25% midterm
35% final.

Letter grades are assigned at the end of the semester ranging from A+ to F.

Copyright, Sources, AI, and Large Language Models

All course materials are copyright protected. Students are free to use them as part of the class. Distribution is only permitted with instructor approval. Uploading materials on webshare sites (e.g., CourseHero) is explicitly prohibited.

The use of all sources used in homework, and project have to be explicitly declared.

AI and Large Language Models (e.g., ChatGPT, BERT, PaLM, Gemini, T5, Turing NLG....) is not permitted unless explicitly stated. All parts of LLMs need to be explicitly declared as sources and all parts have to be explicitly marked as direct quotations.

Homework

Homework is to be handed in via gradescope.

The lowest homework grade will be dropped with one exception: One of the homeworks will be a literature review; this homework cannot be the one you are dropping.

Homework due dates

Homework is expected to be submitted by the time indicated on the homework sheet. When submitted within 2h of the indicated due date, no points are deducted. When submitting 24h or more after the due date, your homework is still graded, but zero points are awarded. In the time window between 2 and 24h after the due date the maximum obtainable points are gradually reduced commensurate to the time passed since the due date. Each student has two 24h extensions to homeworks throughout the semester. Students may use those extensions at any time and are automatically applied to the first two times homework is submitted past the 2h courtesy period and within 24h of the original due date and time (excluding the 2h courtesy period). NOTE: Always go off the homework sheet and not the gradescope time; to accomodate the variable 24h period, it might appear like there is an extension on gradescope, although you cannot take an extra 24h on more than two assignments in the semester.

Exams

This class has a midterm and a final exam.

Boilermaker Pledge and University Resources

As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue.

Nondiscrimination Statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person and encourages each individual to strive to reach their

own potentia in pursuit of its goal of academic excellence. Act accordingly!

Mental Health
If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS). TAO is available to all students at any time by creating an account on the TAO Connect website, or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression and other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 a.m.- 5 p.m.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect. Students in Indianapolis will find support services curated on the Vice Provost for Student Life website. If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette or Indianapolis.

Academic Integrity
Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breeches of this value by either emailing `integrity@purdue.edu` or by calling 765-494-8778. While information may be submitted anonymously, the more information that is submitted provides the greatest opportunity for the university to investigate the concern.

Emergency Preparedness and Readiness
See the separate document posted on Brightspace in the Emergency Preparedness and Readiness folder.